

5.12 Classwork: Trigonometric Transformations & Derivatives — Markscheme

Maximum mark: 8 Ferris wheel with diameter 80 m, lowest point 5 m above ground, period 16 minutes.

$$h(t) = a \cos(bt) + d.$$

- (a) Find a , b , d . [4]

$$\text{Radius} = 40, \text{ centre height} = 5 + 40 = 45 \quad (\mathbf{A1})$$

$$d = 45 \quad (\mathbf{A1})$$

$$a = -40 \quad (\text{negative: starts at minimum with cosine}) \quad (\mathbf{A1})$$

$$\text{Period} = 16, \text{ so } \frac{2\pi}{b} = 16 \Rightarrow b = \frac{\pi}{8} \quad (\mathbf{A1})$$

- (b) Find $h(4)$. [2]

$$h(4) = -40 \cos\left(\frac{\pi}{8} \cdot 4\right) + 45 \quad (\mathbf{M1})$$

$$= -40 \cos\left(\frac{\pi}{2}\right) + 45 = -40(0) + 45 = 45 \text{ m} \quad (\mathbf{A1})$$

The seat is at the centre height (moving through the midpoint going up).

- (c) Find $h'(t)$ and state what it represents. [2]

$$h'(t) = -40 \cdot \left(-\frac{\pi}{8}\right) \sin\left(\frac{\pi t}{8}\right) = 5\pi \sin\left(\frac{\pi t}{8}\right) \quad (\mathbf{A1})$$

This represents the rate of change of height (vertical speed) in m/min. (\mathbf{A1})

Maximum mark: 10 $y = 3 \sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 1$

- (a) Describe the transformations: [4]

(i) Vertical stretch by factor 3. (\mathbf{A1})

(ii) Horizontal compression by factor $\frac{1}{2}$. (\mathbf{A1})

(iii) Horizontal translation $\frac{\pi}{4}$ units to the right. (\mathbf{A1})

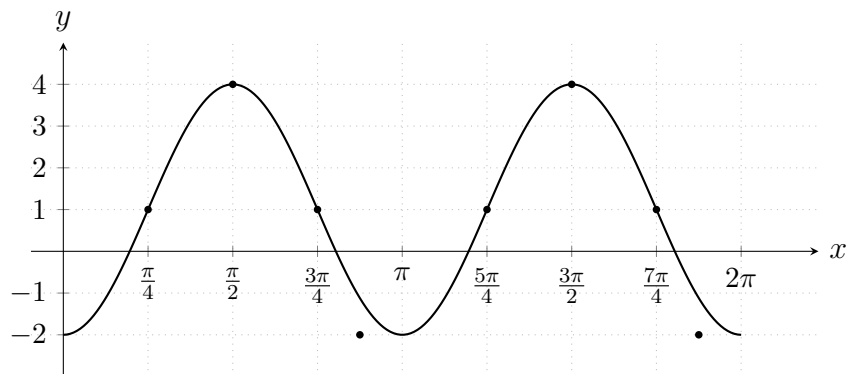
(iv) Vertical translation 1 unit up. (\mathbf{A1})

- (b) Write down: [2]

(i) Amplitude = 3 (\mathbf{A1})

(ii) Period = $\frac{2\pi}{2} = \pi$ (\mathbf{A1})

- (c) Sketch the graph. [4]



Award **(A1)** for correct shape (sinusoidal, 2 complete cycles in $[0, 2\pi]$). **(A1)** for correct amplitude ($\max = 4$, $\min = -2$). **(A1)** for correct horizontal shift (first “mid-crossing” at $x = \pi/4$). **(A1)** for correct vertical shift (midline at $y = 1$).

- Maximum mark: 7 (a) $f(x) = 3 \sin x + 2 \cos x$ [2]
 $f'(x) = 3 \cos x - 2 \sin x$ (A1)(A1)
Award (A1) for each correct term.
- (b) $g(x) = 4 \cos(3x)$ [2]
 $g'(x) = 4 \cdot (-\sin(3x)) \cdot 3 = -12 \sin(3x)$ (M1)(A1)
Award (M1) for applying the chain rule. (A1) for correct answer.
- (c) $h(x) = \sin^2 x = (\sin x)^2$ [3]
 $h'(x) = 2 \sin x \cdot \cos x$ (M1)(A1)
 $= \sin 2x$ (A1)
Award (M1) for chain rule with $2(\sin x)$ as outer derivative. (A1) for $\cdot \cos x$ as inner derivative. Final (A1) for simplified form $\sin 2x$. Accept $2 \sin x \cos x$.

- Maximum mark: 7 $f(x) = 2 \sin(3x) - 1$
- (a) Find $f'(x)$. [2]
 $f'(x) = 2 \cdot 3 \cos(3x) = 6 \cos(3x)$ (M1)(A1)
- (b) Find x -coordinates of max and min. [3]
 $f'(x) = 0 \Rightarrow \cos(3x) = 0$ (M1)
 $3x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \Rightarrow x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$ (A1)
 Maximum at $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$; minimum at $x = \frac{\pi}{2}$ (A1)
Award final (A1) for correctly identifying which are maxima and which is a minimum.
- (c) Find max and min values. [2]
 $f\left(\frac{\pi}{6}\right) = 2 \sin\left(\frac{\pi}{2}\right) - 1 = 2(1) - 1 = 1$ (A1)
 $f\left(\frac{\pi}{2}\right) = 2 \sin\left(\frac{3\pi}{2}\right) - 1 = 2(-1) - 1 = -3$ (A1)
Maximum value is 1, minimum value is -3.

Maximum mark: 6 $g(x) = a \sin(bx) + c$, $\max = 7$, $\min = 1$, period $= \frac{2\pi}{3}$.

(a) Find a , b , c . [3]

$$\text{Amplitude: } a = \frac{7-1}{2} = 3 \quad (\mathbf{A1})$$

$$\text{Vertical shift: } c = \frac{7+1}{2} = 4 \quad (\mathbf{A1})$$

$$\text{Period: } \frac{2\pi}{b} = \frac{2\pi}{3} \Rightarrow b = 3 \quad (\mathbf{A1})$$

(b) Find $g'\left(\frac{\pi}{6}\right)$. [3]

$$g'(x) = 3 \cdot 3 \cos(3x) = 9 \cos(3x) \quad (\mathbf{M1})(\mathbf{A1})$$

$$g'\left(\frac{\pi}{6}\right) = 9 \cos\left(\frac{\pi}{2}\right) = 9(0) = 0 \quad (\mathbf{A1})$$

Maximum mark: 6 $f(x) = \cos 2x + 2 \sin x$

(a) Show that $f'(x) = 2 \cos x(1 - 2 \sin x)$. [3]

$$f'(x) = -2 \sin 2x + 2 \cos x \quad (\mathbf{A1})$$

$$= -2(2 \sin x \cos x) + 2 \cos x \quad (\mathbf{M1})$$

$$= -4 \sin x \cos x + 2 \cos x = 2 \cos x(1 - 2 \sin x) \quad (\mathbf{AG})$$

*Award **(A1)** for correct differentiation. **(M1)** for using $\sin 2x = 2 \sin x \cos x$ and factoring. Must see all steps for **(AG)**.*

(b) Find values of x where $f'(x) = 0$. [3]

$$\cos x = 0 \quad \Rightarrow \quad x = \frac{\pi}{2}, \frac{3\pi}{2} \quad (\mathbf{A1})$$

$$1 - 2 \sin x = 0 \quad \Rightarrow \quad \sin x = \frac{1}{2} \quad (\mathbf{M1})$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6} \quad (\mathbf{A1})$$

All four values required for full marks: $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$.